

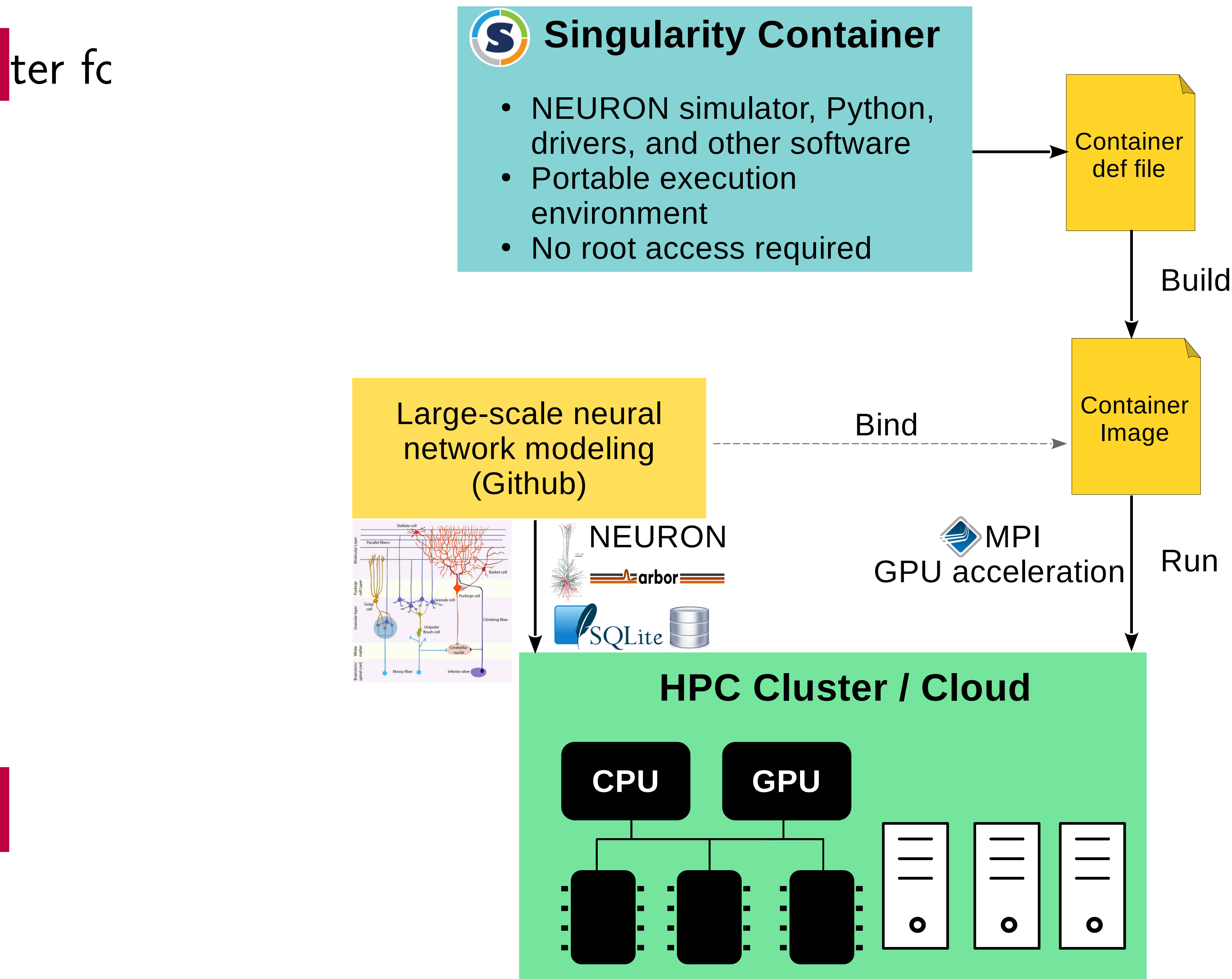
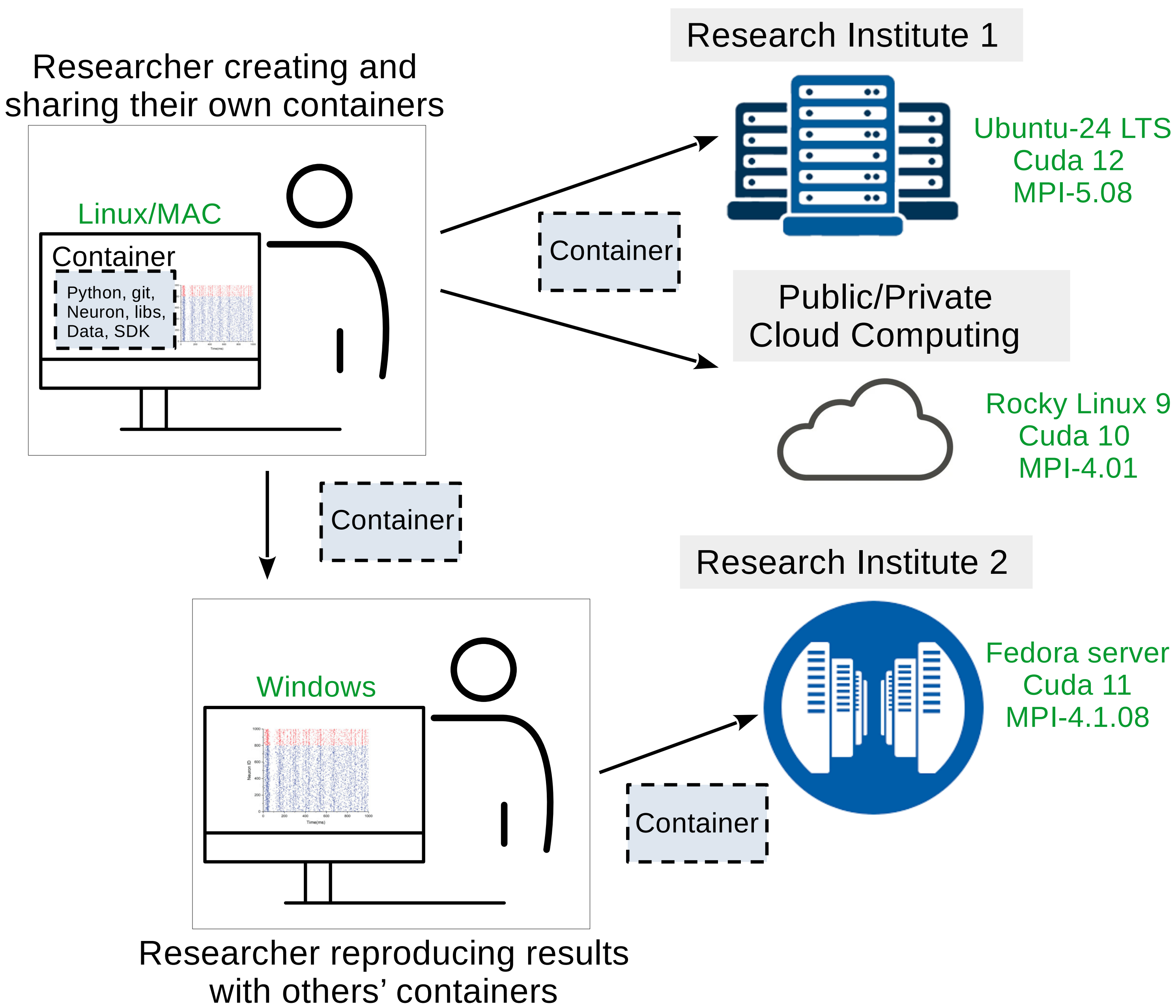
Container-based High-Performance Computing Our Computational Framework for Large-Scale Biological Neural Networks

Introduction

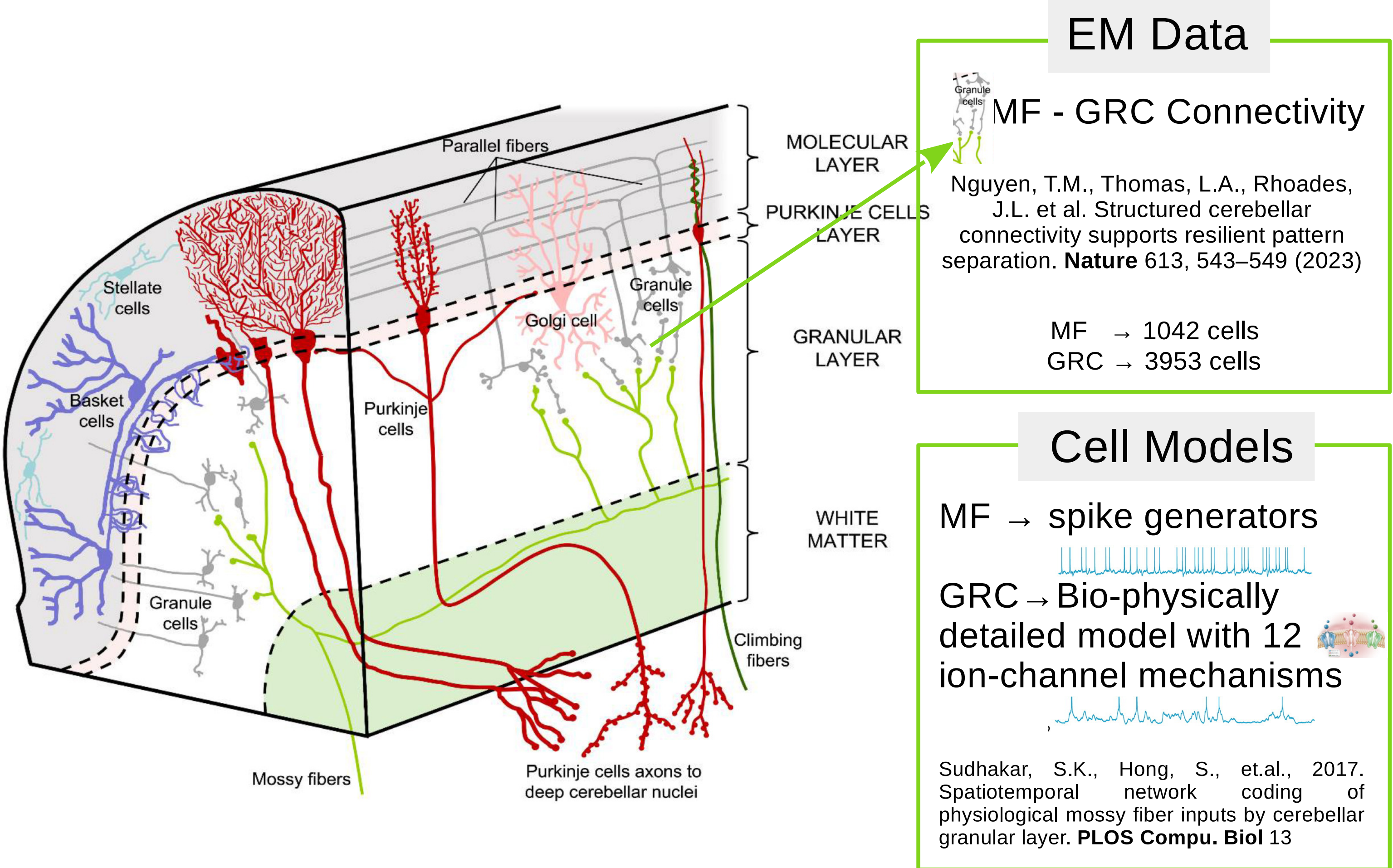
1. Modeling **large-scale, biophysically detailed neural networks** increasingly relies on **high-performance computing (HPC)**.
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to **software incompatibilities** and **complex dependencies**.
3. We address this issue using **Singularity containers** that encapsulates the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.
4. We give practical demonstrations by benchmarking HPC environments with **cerebellar mossy fiber–granule cell network** simulations.

Containerization for Scientific Computing

A container is a lightweight, standalone, virtualized executable environment that includes all dependencies required to run large-scale scientific models.



Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



We run multiple copies ($n = 280$) of the network connectivity model simultaneously to simulate extreme computational workloads. This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

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Benchmarking HPC Environments

Table: Network Configuration

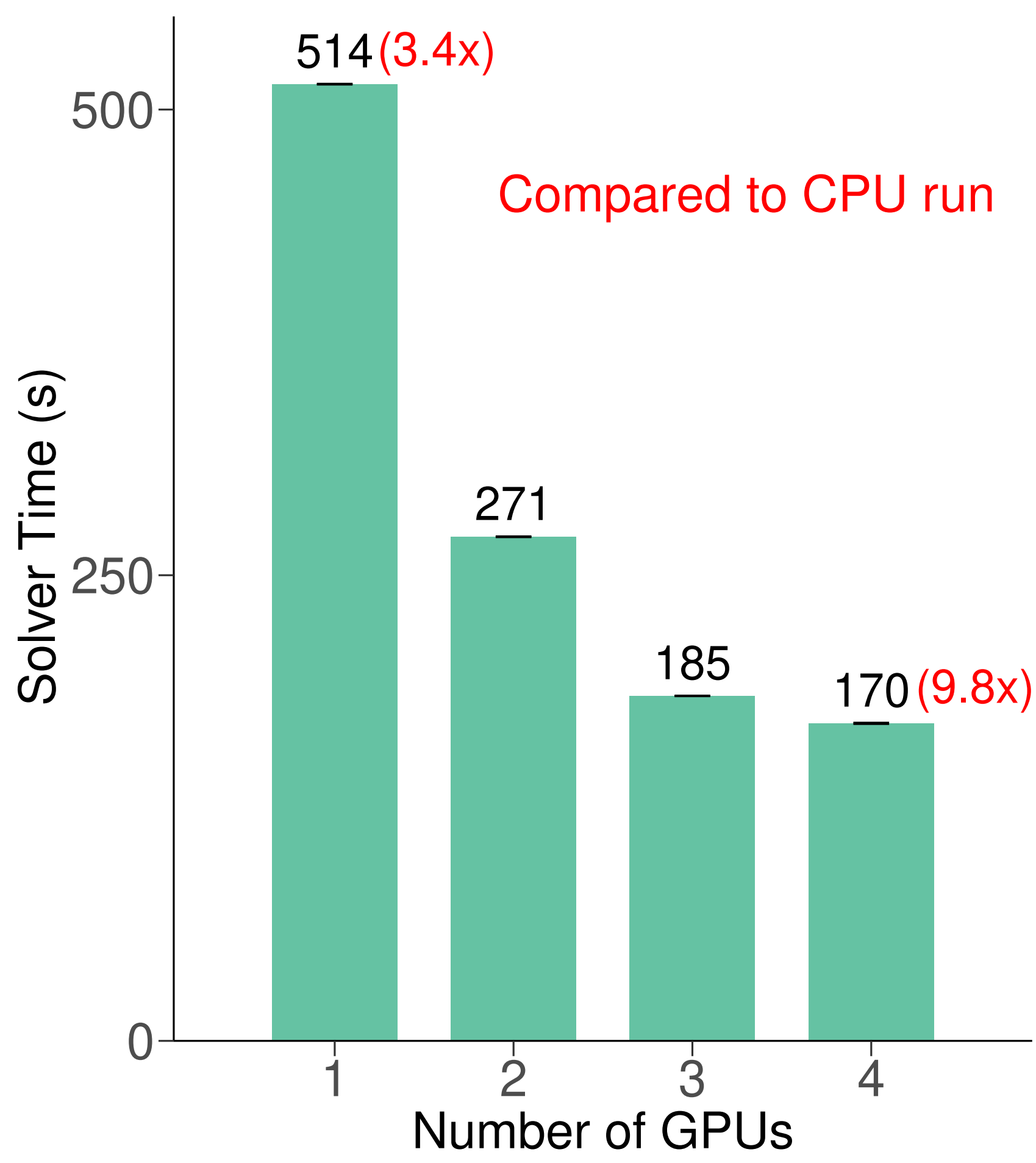
Network elements	Count (millions)
MF	0.299
GRC	1.099
Compartments	3.297
Synapses	3.468

Table: Simulation Setup

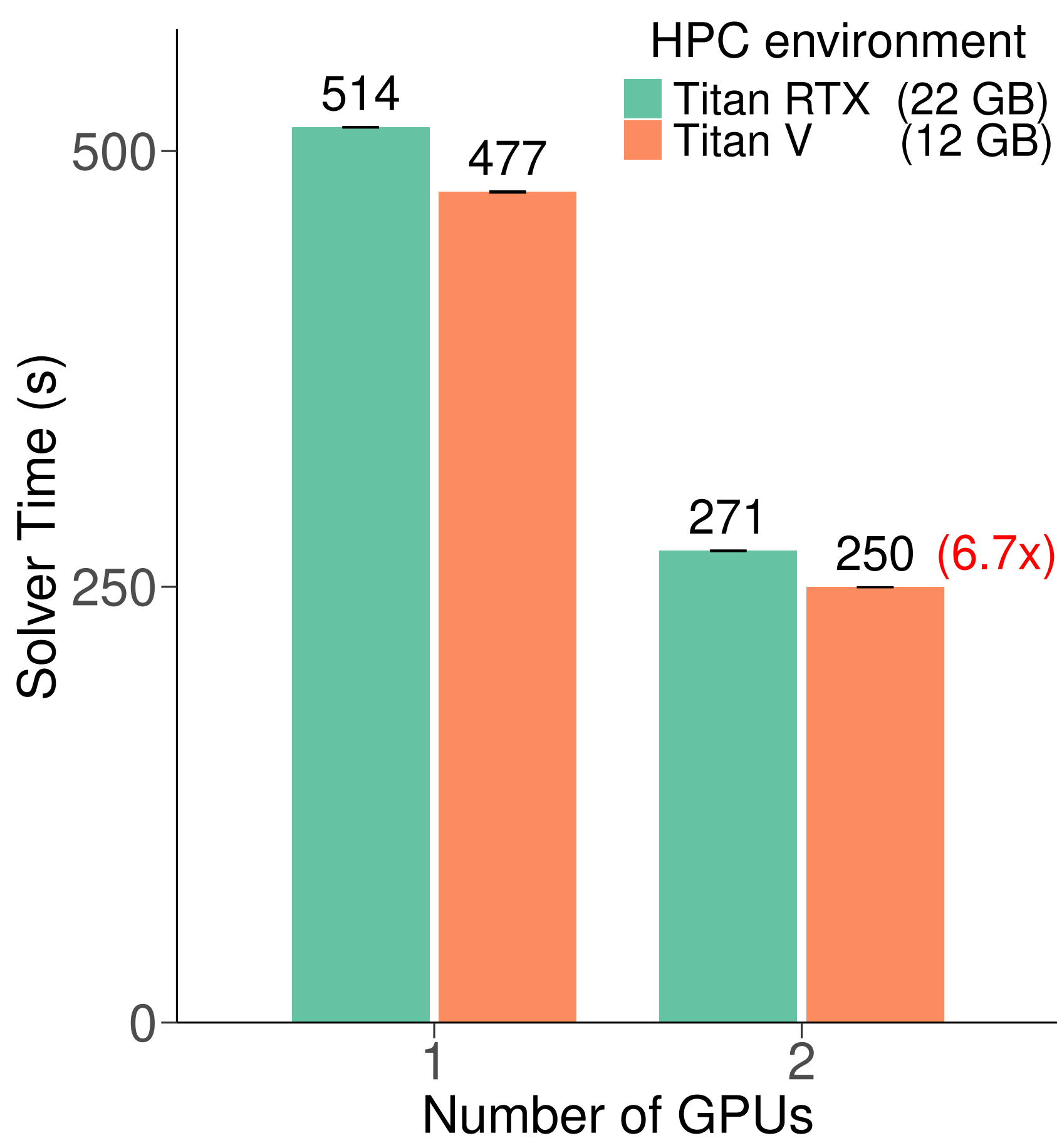
Simulation time	1 s
Simulation software	CoreNEURON

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	1670	CoreNEURON-CPU



(a) TITAN RTX architecture: Solver time vs. varying numbers of GPUs in CORENeuron-GPU simulator.



(b) Comparison of solver time between HPC environments

TITAN RTXs outperform 30-core CPUs by up to 9.8x, with performance improving as GPU count increases.

Future works:

- 🔧 **Develop a biologically realistic cerebellar circuit model** including granule layer (with Golgi cells), Purkinje layer, and other interneurons.
- 🔧 **Benchmark network performance across multiple institutions** to evaluate and understand the scalability and performance limits of HPC systems.

Acknowledgment

This work was supported by IBS-R001-D2-2025-a004D5

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